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MICHAEL LUO

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1 How to solve recurrences

Example 1.1 (Binet's formula). Let F be the Fibonacci sequence satisfying $F_0 = 0$, $F_1 = 1$, and $F_n = F_{n-1} + F_{n-2}$. Show

$$F_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right).$$

We could prove this formula using induction or something similar, but we will instead show how this was derived.

Our approach is to first ignore the initial conditions $F_0 = 0$ and $F_1 = 1$ and instead focus on characterizing *all* sequences a that satisfy the condition

$$a_n = a_{n-1} + a_{n-2}. \quad (\star)$$

Observe that the set of sequences satisfying $(\star)$ is closed under *linear combinations*. That is, if sequences u and v satisfy $(\star)$, for any real numbers a and b , $au + bv$ satisfies $(\star)$ as well, where scalar multiplication and addition are defined component-wise. Buzzword: the set of sequences satisfying the Fibonacci relation forms a vector space with dimension 2.

The big leap here is to guess two “nice” sequences u and v that satisfy the Fibonacci relation. Then, we hope we can find a and b such that $au + bv$ has zeroth term 0 and first term 1, which gives the Fibonacci sequence. We guess the geometric sequences $u_n = \alpha^n$ and $v_n = \beta^n$ for some α and β . Then the Fibonacci relations become

$$\begin{aligned} u_n = u_{n-1} + u_{n-2} &\iff \alpha^n = \alpha^{n-1} + \alpha^{n-2} \iff \alpha^2 - \alpha - 1 = 0 \\ v_n = v_{n-1} + v_{n-2} &\iff \beta^n = \beta^{n-1} + \beta^{n-2} \iff \beta^2 - \beta - 1 = 0. \end{aligned}$$

So α and β are roots of the polynomial $P(\lambda) = \lambda^2 - \lambda - 1$. We call P the *characteristic polynomial* of the recurrence relation $a_n - a_{n-1} - a_{n-2} = 0$. It's not a coincidence that the coefficients of the characteristic polynomial correspond with the coefficients of the recurrence relation. Similarly, the characteristic polynomial of the recurrence relation $a_n = 3a_{n-1} - 2a_{n-3}$ (or $a_n - 3a_{n-1} + 2a_{n-3} = 0$) is $\lambda^3 - 3\lambda^2 + 2$.

Exercise 1.2. What is the characteristic polynomial of the recurrence relation $a_n = 4a_{n-2} - a_{n-3}$?

The roots of the characteristic polynomial correspond to the geometric sequences that satisfy the recurrence relation. Going back to the Fibonacci sequence, the roots of the characteristic polynomial $\lambda^2 - \lambda - 1$ are $\alpha = \frac{1+\sqrt{5}}{2}$ and $\beta = \frac{1-\sqrt{5}}{2}$ from the quadratic formula. Therefore $u_n = \alpha^n$ and $v_n = \beta^n$.

Now we want to find a and b such that the initial conditions are satisfied. In other words, we want

$$\begin{aligned} au_0 + bv_0 &= F_0 \\ au_1 + bv_1 &= F_1. \end{aligned}$$

The system of equations becomes

$$\begin{aligned} a + b &= 0 \\ a \left(\frac{1 + \sqrt{5}}{2} \right) + b \left(\frac{1 - \sqrt{5}}{2} \right) &= 1. \end{aligned}$$

Solving, we see $a = \frac{1}{\sqrt{5}}$ and $b = -\frac{1}{\sqrt{5}}$, so

$$F_n = au_n + bv_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right),$$

as desired.

Exercise 1.3. If $a_0 = 2$, $a_1 = 4$, and $a_n = a_{n-1} + 12a_{n-2}$, find an explicit formula for a_n .

2 Problems

Problem 2.1. If $a_0 = 0$ and $a_n = 3a_{n-1} + 1$ for $n \geq 1$, find a_n .

Problem 2.2 (HMMT 2012/Guts 2). Let $a_0, a_1, a_2, \dots$ denote the sequence of real numbers such that $a_0 = 2$ and $a_{n+1} = \frac{a_n}{1+a_n}$ for $n \geq 0$. Compute a_{2012} .

Problem 2.3. Find the number of ways to color each cell of a 2×6 grid black or white such that no 2×2 square is completely black.

Problem 2.4 (CMIMC 2023/A Estimation). Consider the sequence given by $x_1 = 1$ and $x_{n+1} = 1 + \frac{1}{x_n}$ for $n \geq 1$. As n grows large, x_n gets closer and closer to $\varphi = \frac{1+\sqrt{5}}{2}$. Approximate $\log_{1/2} |x_{2023} - \varphi|$.

Problem 2.5 (SMT 2024/T10). A group of scientists are researching the population dynamics of a certain bacterial colony. They find that if P_n represents the bacterial population at n minutes from the start, then this population can be modeled in a surprisingly discrete way given by $P_n = \frac{P_{n-1}^3}{P_{n-3}^4}$. Given that in one experiment, $P_0 = 1$, $P_1 = 1$, and $P_2 = 7^3$, find $3 \log_7(P_{100})$.

Problem 2.6 (CMIMC 2024/A9). Let $\mathbb{Q}_{\geq 0}$ be the nonnegative rational numbers, $f : \mathbb{Q}_{\geq 0} \rightarrow \mathbb{Q}_{\geq 0}$ such that $f(z+1) = f(z) + 1$, $f(1/z) = f(z)$ for $z \neq 0$, and $f(0) = 0$. Define a sequence P_n of nonnegative integers recursively via $P_0 = 0$, $P_1 = 1$, and $P_n = 2P_{n-1} + P_{n-2}$ for every $n \geq 2$. Find $f\left(\frac{P_{20}}{P_{24}}\right)$.

3 Solutions

Solution to Problem 2.1. We can't write down the characteristic polynomial for this sequence immediately because of the pesky $+1$ term. Therefore, we are motivated to *shift the sequence* to cancel out the $+1$ term.

Define the sequence b such that $a_n = b_n + c$ for some constant c . Now the condition becomes $b_n + c = 3(b_{n-1} + c) + 1$, which rearranges to $b_n = 3b_{n-1} + 2c + 1$. Since we want to yeet (cancel out) the constant term, we set $c = -\frac{1}{2}$. Now $b_0 = a_0 + \frac{1}{2} = \frac{1}{2}$ and $b_n = 3b_{n-1}$. This is a geometric sequence, so $b_n = \frac{3^n}{2}$, and $a_n = \frac{3^n}{2} - \frac{1}{2}$. ■

Solution to Problem 2.2. Computing the first few terms $a_0 = \frac{2}{1}$, $a_1 = \frac{2}{3}$, $a_2 = \frac{2}{5}$, and $a_3 = \frac{2}{7}$, it's pretty clear $a_n = \frac{2}{2n+1}$, so $a_{2012} = \frac{2}{4025}$. You could also prove this by induction or whatever. ■

Solution to Problem 2.3. Casework on this thing would be absolutely disgusting, so we use recursion. Call a coloring *good* if it contains no 2×2 blocks that are completely black. Call a coloring *xooks* if its rightmost edge is black-black, and *rbo* otherwise. Let a_n be the number of good xooks colorings of a $2 \times n$ grid, and let b_n be the number of good rbo colorings of a $2 \times n$ grid. We want $a_6 + b_6$.

We observe $a_n = b_{n-1}$ since you can add a black-black edge to any rbo $2 \times (n-1)$ grid to make a xooks $2 \times n$ grid, and $b_n = 3(a_{n-1} + b_{n-1})$ since you can add a white-white, white-black, or black-white edge to any xooks or rbo $2 \times (n-1)$ grid to make an rbo $2 \times n$ grid.

Our initial conditions are $a_0 = 0$ and $b_0 = 1$. We compute everything recursively as shown below.

n	0	1	2	3	4	5	6
a_n	0	1	3	12	45	171	648
b_n	1	3	12	45	171	648	2457

Our answer is $a_6 + b_6 = 648 + 2457 = 3105$. ■

Solution to Problem 2.4. Let $n = 2023$. By induction or something, we observe $x_n = \frac{F_{n+1}}{F_n}$. So by Binet,

$$x_n - \varphi = \frac{F_{n+1}}{F_n} - \varphi = \frac{\varphi^{n+1} - \varphi^{-n-1}}{\varphi^n - \varphi^{-n}} - \varphi = \frac{\varphi^{2n+2} - 1}{\varphi^{2n+1} - \varphi} - \varphi = \frac{\varphi^2 - 1}{\varphi^{2n+1} - \varphi}.$$

Since we are estimating the logarithm, we basically drop everything:

$$\log_{1/2} \left(\frac{\varphi^2 - 1}{\varphi^{2n+1} - \varphi} \right) \approx \log_{1/2} \left(\frac{1}{\varphi^{2n+1}} \right) = 4047 \log_2 \varphi \approx 4000 \log_2 1.62 = 4000 \frac{\log 1.62}{\log 2}.$$

Recall from Fermi Questions $\log 2 \approx 0.3$ and $\log 3 \approx 0.48$, so $\log 1.62 = \log 162 - 2 = \log 2 + 4 \log 3 - 2 \approx 0.22$ and $\log 2 \approx 0.3$, so

$$4000 \frac{\log 1.62}{\log 2} \approx 4000 \frac{0.22}{0.30} \approx 2933.$$

Not too far from the actual answer of 2808-ish. ■

Solution to Problem 2.5. Okay what are we even doing here with the base 7 logarithms. We yeet (get rid of) them immediately by defining $Q_n = \log_7 P_n$. Now $Q_n = 3Q_{n-1} -$

$4Q_{n-3}$, and $Q_0 = Q_1 = 0$ and $Q_2 = 3$. The characteristic polynomial is $\lambda^3 - 3\lambda^2 + 4$, which factors into $(\lambda + 1)(\lambda - 2)^2$.

Since the roots of the characteristic polynomial are -1 and 2 , $Q_n = a(-1)^n + b2^n$ for some real a, b . Plugging in $n = 0$ gives $a + b = 0$, plugging in $n = 1$ gives $-a + 2b = 0$, and plugging in $n = 2$ gives $a + 4b = 3$. But oh no! This linear system of two variables and three equations has no solutions.

The problem is that our characteristic polynomial had a double root, so we only had two “different” sequences, which gave us a 2 variable, 3 equation system. We need to somehow find another sequence that works. Miraculously, the sequence $n2^n$ works. (Check it! Similarly, if 2 was a triple root, the sequence n^22^n would also work.) Therefore, we have $Q_n = a(-1)^n + b2^n + cn2^n$ for some a, b, c . From plugging in $n = 0, 1, 2$, we have the system

$$\begin{aligned} a + b &= 0 \\ -a + 2b + 2c &= 0 \\ a + 4b + 8c &= 3. \end{aligned}$$

Solving, we see $(a, b, c) = (\frac{1}{3}, -\frac{1}{3}, \frac{1}{2})$, so $Q_n = \frac{1}{3}(-1)^n - \frac{1}{3} \cdot 2^n + \frac{1}{2}n \cdot 2^n$, and

$$3Q_{100} = (-1)^{100} - 2^{100} + \frac{3}{2} \cdot 100 \cdot 2^{100} = 149 \cdot 2^{100} + 1. \quad \blacksquare$$

Solution to Problem 2.6. We only need to know about P_n when n is a multiple of 4, so we are motivated to define the sequence $Q_n = P_{4n}$. The characteristic polynomial of P is $\lambda^2 - 2\lambda - 1$, which has roots $1 + \sqrt{2}$ and $1 - \sqrt{2}$. Therefore, there exist a and b such that

$$P_n = a(1 + \sqrt{2})^n + b(1 - \sqrt{2})^n.$$

Then,

$$Q_n = P_{4n} = a(1 + \sqrt{2})^{4n} + b(1 - \sqrt{2})^{4n} = a(17 + 12\sqrt{2})^n + b(17 - 12\sqrt{2})^n.$$

Now we turn this explicit formula *back into a recurrence relation*. The roots of the characteristic polynomial of Q are $17 + 12\sqrt{2}$ and $17 - 12\sqrt{2}$, so by Vieta's formulas, it is $\lambda^2 - 34\lambda + 1$, so $Q_n = 34Q_{n-1} - Q_{n-2}$.

After playing around with certain values of z , we observe $f(\frac{p}{q})$ is the number of steps in the Euclidean algorithm reducing p and q until one of them is zero. But if $p < q$, the pairs (p, q) and $(q - p, q)$ become $(p, q - p)$ after one step, so (p, q) and $(q - p, q)$ take an equal number of steps to reduce to zero, so $f(\frac{p}{q}) = f(\frac{q-p}{q})$.

Now we compute

$$f\left(\frac{Q_5}{Q_6}\right) = f\left(\frac{Q_6}{Q_5}\right) = f\left(\frac{34Q_5 - Q_4}{Q_5}\right) = 33 + f\left(\frac{Q_5 - Q_4}{Q_5}\right) = 33 + f\left(\frac{Q_4}{Q_5}\right).$$

Similarly, we have

$$\begin{aligned} 33 + f\left(\frac{Q_4}{Q_5}\right) &= 33 \cdot 2 + f\left(\frac{Q_3}{Q_4}\right) = \dots = 33 \cdot 4 + f\left(\frac{Q_1}{Q_2}\right) \\ &= 33 \cdot 4 + f\left(\frac{12}{34 \cdot 12}\right) = 33 \cdot 4 + 34 = 166. \quad \blacksquare \end{aligned}$$